COMPLETIONS OF A SUBMODULE AND OF A QUOTIENT, AND THE CLOSURE OF THE IMAGE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Completions as inverse limits over a fundamental system of open submodules, and exactness of inverse limits of short exact sequences. Standard, used at proof lines 959-967. Reduces exactness of the completions to exactness of the quotients.
- The canonical map from a module to its completion has dense image, and the kernel of the induced map on completions is closed. Standard, used at proof lines 967-969. Identifies the completion of the submodule with the closure of the image.

Lemma 0.1. *Let R be a topological ring. Let M be a linearly topologized R-module. Let $N \subset M$ be a submodule. Then*

(1) $0 \to N^\wedge \to M^\wedge \to (M/N)^\wedge$ *is exact, and*
(2) $N^\wedge$ *is the closure of the image of* $N \to M^\wedge$.

Proof. Let M_λ, $\lambda \in \Lambda$ be a fundamental system of open submodules. Then $N \cap M_\lambda$ is a fundamental system of open submodules of N and $M_\lambda + N/N$ is a fundamental system of open submodules of M/N. Thus we see that (1) follows from the exactness of the sequences

$$0 \to N/N \cap M_\lambda \to M/M_\lambda \to M/(M_\lambda + N) \to 0$$

and the fact that taking limits commutes with limits. The second statement follows from this and the fact that $N \to N^\wedge$ has dense image and that the kernel of $M^\wedge \to (M/N)^\wedge$ is closed. □

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